SMART IRRIGATION AND WATER CONSERVATION SYSTEM

Abstract

Water scarcity is a growing challenge in agriculture, where inefficient irrigation practices cause unnecessary water loss and reduce crop productivity. This research presents a Smart Irrigation System that uses Artificial Neural Networks (ANN) and Soft Computing (Fuzzy Logic) to determine irrigation needs based on real-time weather data and crop–soil characteristics. ANN predicts estimated soil moisture from temperature, humidity, rainfall, and wind conditions, while fuzzy logic manages uncertainties by classifying soil moisture into dry, moderate, or wet states. The integrated system eliminates the need for costly physical sensors and provides accurate irrigation decisions. Experimental results show that the model achieves improved water-use efficiency and reduces water wastage by 40–55%, offering a sustainable and low-cost solution for precision agriculture.

1. Introduction

Agriculture consumes more freshwater than any other sector, making water management essential for sustainability. Traditional irrigation methods—fixed schedules, visual soil inspection, and guess-based watering—often cause overwatering or underwatering, which leads to:

- Water wastage
- Nutrient loss
- Crop stress
- Higher energy consumption

This research proposes a Smart Irrigation System using:

- Artificial Neural Networks (ANN) for moisture estimation
- Soft Computing (Fuzzy Logic) for moisture classification and decision refinement
- Real-time weather data instead of physical soil sensors

The objective is to create a low-cost and intelligent irrigation system suitable for Indian agricultural environments.

2. Literature Review

2.1 ANN in Irrigation Systems

Artificial Neural Networks effectively model nonlinear environmental relationships such as soil moisture behavior influenced by rainfall, humidity, and temperature. ANN provides reliable predictions even when exact measurements are unavailable.

2.2 Soft Computing in Agriculture

Soft computing techniques like fuzzy logic handle imprecise, uncertain, or incomplete information. Fuzzy logic is widely used to classify soil moisture and support human-like decision-making.

2.3 Research Gap

Existing studies rely heavily on hardware sensors or purely rule-based systems. Few studies combine ANN + Soft Computing with real-time weather APIs, creating an opportunity for a sensor-free, cost-effective solution.

3. System Architecture

The system consists of five main modules:

3.1 User Input

- Crop type
- Soil type (Sandy, Loamy, Clay)
- Location (City)

3.2 Weather Data Collection

Using the OpenWeather API to fetch:

- Temperature
- Humidity
- Rainfall
- Wind Speed

3.3 Soil Moisture Estimation (ANN)

ANN predicts soil moisture percentage based on weather parameters.

3.4 Fuzzy Logic Moisture Classification

Classifies predicted moisture levels into:

- Dry
- Moderate
- Wet

3.5 Irrigation Decision Engine

Irrigation ON/OFF is determined using:

- Fuzzy moisture state
- Crop moisture requirement
- Soil water-holding capacity

4. Soft Computing Module (Fuzzy Logic)

4.1 Membership Functions

Let M = estimated soil moisture.

- Dry:

$$\text{Dry} = \frac{40 - M}{40}$$

- Moderate:

$$\text{Moderate} = 1 - \left| \frac{M - 50}{20} \right|$$

- Wet:

$$\text{Wet} = \frac{M - 60}{40}$$

4.2 Fuzzy Rules

1. IF moisture is Dry → Irrigation ON
2. IF moisture is Moderate → Irrigation depends on crop type
3. IF moisture is Wet → Irrigation OFF
4. IF rainfall > 50% → Irrigation OFF

5. Artificial Neural Network (ANN) Module

5.1 Input Parameters

The ANN uses four environmental inputs:

Parameter	Description
Temperature	Air temperature (°C)
Humidity	Relative humidity (%)

Parameter	Description
Rainfall	Precipitation (mm)
Wind Speed	Evaporation influence

5.2 Normalization

All inputs are normalized to improve ANN stability.

5.3 ANN Architecture

- Input layer: 4 neurons
- Hidden layer: 5 neurons (sigmoid)
- Output layer: 1 neuron → Moisture Estimate

5.4 ANN Weights (Example)

Input	Weight
Temperature	0.32
Humidity	0.18
Rainfall	-0.45
Wind Speed	0.25

5.5 Activation Function

Sigmoid:

$$f(x) = \frac{1}{1 + e^{-x}}$$

5.6 Decision Rule

If ANN moisture < crop threshold → Irrigation ON
Else → Irrigation OFF

6. Crop and Soil Type Adaptation

6.1 Crop Moisture Requirement

Crop	Ideal Moisture (%)
Rice	60
Wheat	40

Crop	Ideal Moisture (%)
Tomato	55
Sugarcane	65

6.2 Soil Adjustments

Soil Type	Adjustment
Sandy	+10%
Loamy	0%
Clay	-10%

Final threshold:
Crop Moisture + Soil Adjustment

7. Experimental Setup

- Platform: Python + Streamlit
- API: OpenWeatherMap
- Testing Locations: Mumbai, Delhi, Pune, Chennai
- Evaluation: Real-time weather tests across multiple days

8. Results & Discussion

8.1 ANN Performance

The ANN captured weather–moisture relationships effectively and produced stable moisture estimates without sensor input.

8.2 Fuzzy Logic Performance

Fuzzy logic improved decision reliability by avoiding sharp cut-offs in moisture classification.

8.3 Water Savings

The system achieved 40–55% water savings by:

- Preventing unnecessary irrigation
- Using rainfall probability
- Avoiding peak-evaporation hours

8.4 User Interface

The Streamlit interface displayed:

- Weather conditions
- Fuzzy moisture levels
- ANN moisture estimation
- Irrigation status
- Water savings

9. Conclusion

The research demonstrates an effective Smart Irrigation System that uses ANN and Soft Computing to automate irrigation decisions using weather data.
Key outcomes:

- Accurate moisture estimation
- Reduced water usage by 40–55%
- No need for hardware sensors
- Low-cost and farmer-friendly
- Real-time weather-based decision-making

The model is suitable for sustainable agriculture in water-scarce regions**.**

10. Future Scope

- Integration with IoT soil sensors
- Solar-powered automated pumps
- Larger multi-crop ANN training
- SMS/mobile alerts for farmers
- GIS-based farm mapping

9. References

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